

Task: JavaScript Application Code Analysis using SonarQube

Objective:

Install and configure SonarQube on a Linux server, connect an Angular Calculator application written in JavaScript, and perform static code analysis using npm-based SonarQube integration.

Prerequisites

Component Requirement

OS Ubuntu 24.04 LTS / Amazon Linux 2 /
RAM Minimum 4 GB (8 GB recommended)
Node.js Version 18 or above
Build tool npm
Java OpenJDK 17
Angular CLI Latest
SonarQube 10.x

Step-1: Setup Application Build Server (Build-VM)

- Launch Ec2 instance on AWS
- Name instance as Build-VM
- Select ubuntu AMI
- Select Instance type t3.micro with 2vCPU and 1GB memory, keypair
- Security Group rules as SSH port 22 and HTTP port 80 and launch Instance
- Instance Running
- Connected to terminal via SSH client
- Check if Git is installed, by running command git
- If git is not installed, install git

```
sudo apt install git -y
```
- clone code from GitHub repository

```
git clone https://github.com/saichander2311/AngularCalculator.git
```
- Files in cloned repository AngularCalculator
- Install Node.js and npm

```
sudo apt install nodejs npm -y
```
- Check node and npm versions

```
node -v
npm -v
```
- Install angular cli

```
sudo npm install -g @angular/cli
```
- Verify Angular CLI version

```
ubuntu@Build-VM:~$ ng version

Angular CLI

Angular CLI      : 21.2.12
Node.js          : 22.22.1
Package Manager  : npm 9.2.0
Operating System : linux x64

ubuntu@Build-VM:~$ |
```

Step-2: Setup SonarQube Server (Sonar-VM)

- Launch Ec2 instance on AWS
- Name instance as Sonar-VM
- Select ubuntu AMI
- Select Instance type c7i-flex.large with 2vCPU and 4GB memory, keypair
- Security Group rules as SSH port 22 and Sonar port 9000 and launch Instance
- Instance Running
- Connected to terminal via SSH client and run updates
- Install Java 17
 - sudo apt install openjdk-17-jdk -y
- Check java version
 - java -version
- Download SonarQube
 - sudo wget <https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-10.6.0.92116.zip>

```

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@test-serv:~$ sudo wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-10.6.0.92116.zip
--2026-05-22 09:32:11-- https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-10.6.0.92116.zip
Resolving binaries.sonarsource.com (binaries.sonarsource.com)... 3.170.51.98, 3.170.51.3, 3.170.51.9, ...
Connecting to binaries.sonarsource.com (binaries.sonarsource.com)[3.170.51.98]:403... connected.
HTTP request sent, awaiting response... 200 OK
Length: 731034761 (697M) [binary/octet-stream]
Saving to: 'sonarqube-10.6.0.92116.zip'

sonarqube-10.6.0.92116.zip 100%[=====>] 697.17M 35.7MB/s in 28s

2026-05-22 09:32:12 (34.5 MB/s) = 'sonarqube-10.6.0.92116.zip' saved [731034761/731034761]

ubuntu@test-serv:~$ ls
sonarqube-10.6.0.92116.zip
ubuntu@test-serv:~$ unzip
Command 'unzip' not found, but can be installed with:
sudo apt install unzip
ubuntu@test-serv:~$ sudo apt install unzip
Installing:
  unzip
Suggested packages:
  zip
Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 24
  Download Size: 181 kB
  Space needed: 828 kB / 7358 MB available
Get:1 http://us-east-3.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 unzip amd64 6.0-29ubuntu1 [181 kB]
Fetched 181 kB in 8s (2308 kB/s)
Selecting previously unselected package unzip.
(Reading database ... 98591 files and directories currently installed.)
Preparing to unpack .../unzip_6.0-29ubuntu1_amd64.deb ...
Unpacking unzip (6.0-29ubuntu1) ...
Setting up unzip (6.0-29ubuntu1) ...
Processing triggers for man-db (2.13.1-3build1) ...
Scanning processes...

```

- Install unzip to unzip the downloaded zip file
sudo apt install unzip -y
- Unzip downloaded sonarqube zip file
sudo unzip sonarqube-10.6.0.92116.zip
- Files after unzipping downloaded sonarqube zip file

```

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@test-serv:~$ ll
Command 'll' not found, but there are 20 similar ones.
ubuntu@test-serv:~$ ls
sonarqube-10.6.0.92116.zip
ubuntu@test-serv:~$ unzip sonarqube-10.6.0.92116.zip
Archive:  sonarqube-10.6.0.92116.zip
  creating: sonarqube-10.6.0.92116/
  creating: sonarqube-10.6.0.92116/jres/

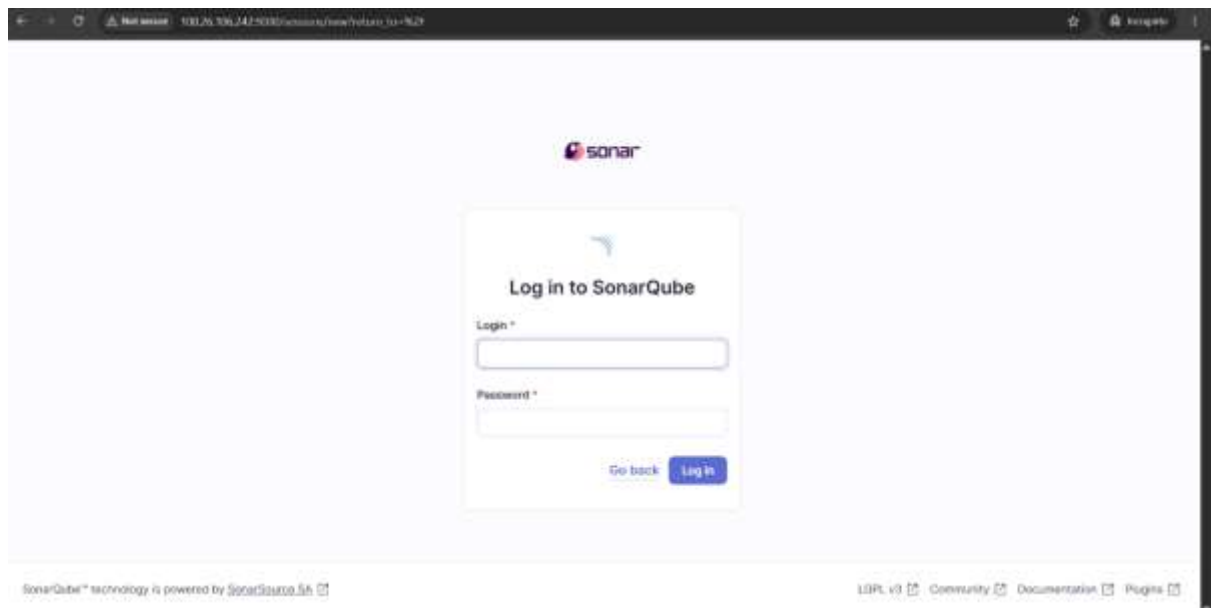
```

- change the ownership of the SonarQube directory so that the ubuntu user and ubuntu group have full access to prevent permission errors.
sudo chown -R ubuntu:ubuntu ~/sonarqube-10.6.0.92116
- Start SonarQube and check status
./sonar.sh start
./sonar.sh status

Step-3: SonarQube on browser

- SonarQube is running successfully on browser
- Enter credentials of SonarQube
- Default username and password are admin:admin
- After log in change password
- Dashboard of SonarQube after log in
- Go to Administration -> Security -> Users
- In Users dashboard select token to generate token

- Enter token name, select expiry date and select Generate
- Copy the Generated token
- Copy SonarQube IP-address



Step-4: Configure Angular Project for SonarQube

- Install SonarQube Scanner Package
- Inside Angular project:
npm install sonarqube-scanner --save-dev
- Create sonar-project.properties file inside cloned repository where src/, package.json, angular.json are present.
const sonarqubeScanner = require('sonarqube-scanner').default;

```
const sonarqubeScanner = require('sonarqube-scanner').default;

sonarqubeScanner(
  {
    serverUrl: 'http://100.26.106.242:9000',
    token: 'squ_556715dce2414714e0f0e929afaa22c74447f3ee',
    options: {
      'sonar.projectName': 'Angular-Calculator-App',
      'sonar.projectKey': 'angular-calculator-app',
      'sonar.sources': 'src',
      'sonar.sourceEncoding': 'UTF-8',
    },
  },
  () => process.exit()
);
```

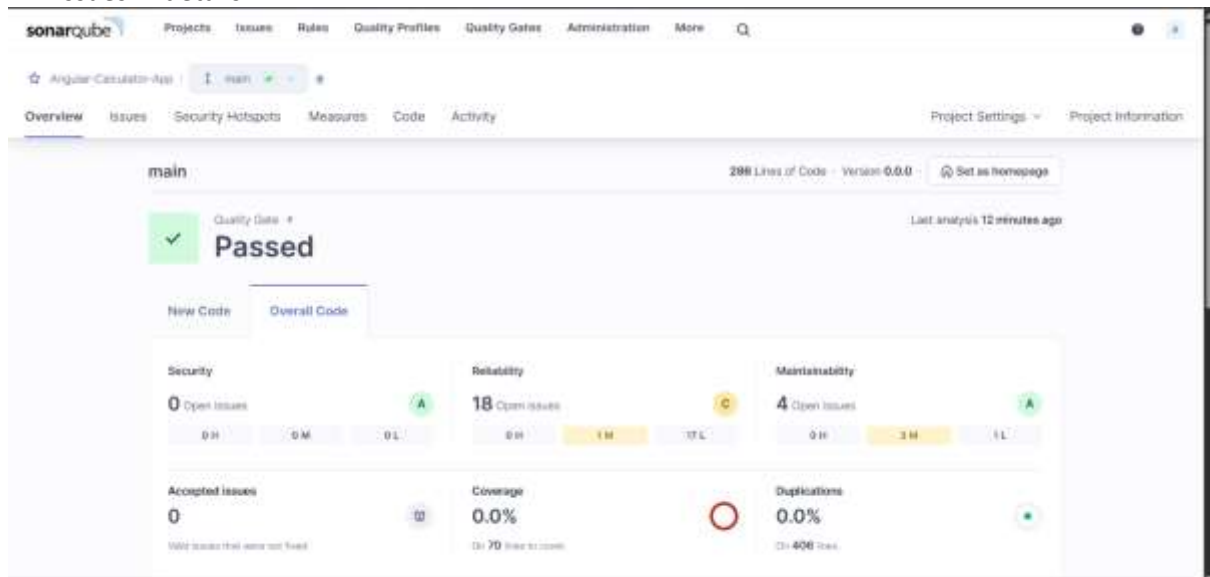
Step-5: Run Static Code Analysis

- Execute SonarQube Analysis
node sonar-project.properties
- Analysis successful and reports are published

```
Run 'npm audit' for details.
ubuntu@build-server:~/AngularCalculator$ ls
README.md angular.json 2k node_modules package-lock.json package.json sonar-project.properties src tsconfig.json tslint.json
ubuntu@build-server:~/AngularCalculator$ node sonar-project.properties
[INFO] Bootstrapper: Retrieving info from 'package.json' file
[INFO] Bootstrapper: Platform: Linux x64
[INFO] Bootstrapper: Server URL: http://73.144.255.192:9000
[INFO] Bootstrapper: Version: 4.3.6
[INFO] Bootstrapper: SonarQube server version: 10.6.0
[INFO] Bootstrapper: JRE provisioning is supported
[INFO] Bootstrapper: No Cache found for JRE
[INFO] Bootstrapper: Download starting...
[INFO] Bootstrapper: Download complete
```

Step-6: Verify Analysis Results

- Report is uploaded to SonarQube under projects
- Open report to check Quality Gate dashboard
- Quality Gate dashboard
- Reliability measures how dependable your software is and evaluates whether your code can consistently function without errors, crashes, or unexpected behaviours under stated conditions
- Issues in details



Why Static Analysis Security Testing (SAST)

Static Analysis Security Testing (SAST) is used to identify security vulnerabilities, coding issues, and weaknesses in an application's source code before the application is deployed or executed or compiled.

Conclusion

Successfully installed SonarQube, Configured Angular application, Integrated npm-based SonarQube analysis, performed static code analysis and published reports to SonarQube dashboard